

NIIS HACKATHON 2026 — OFFICIAL PROBLEM BRIEF

Track: Neurocore / GeoNavX (Location Intelligence & Robotics)

Challenge ID: PS-11

Title: AI-Powered Smart Campus Navigation System

1. Executive Summary & Problem Overview

Sprawling university and institutional campuses encompass multi-acre geographic footprints featuring complex networks of academic faculties, research laboratories, auditoriums, residential hostels, administrative offices, dining messes, sports complexes, and parking amenities. However, standard commercial mapping platforms (such as Google Maps or Apple Maps) fail significantly within educational campus boundaries.

Commercial maps lack granular pedestrian trail networks, offer zero indoor floor-by-floor traversal guidance (e.g., locating an exact lecture room or laboratory on the 3rd floor of a building), ignore temporary campus blockades (such as ongoing construction, festival pavilions, or maintenance repairs), and completely fail to provide barrier-free, accessible routes for individuals with physical mobility constraints.

The objective of **PS-11 (AI-Powered Smart Campus Navigation System)** is to design and develop an intelligent, interactive, map-based indoor-outdoor navigation platform. The system must deliver context-aware shortest-path computation, multi-floor indoor routing, wheelchair-accessible and obstacle-free pathways, landmark-centric visual directions, and live institutional event awareness for students, faculty, and visiting campus guests.

2. Core Personas & Access Roles

1. **New Students & Campus Visitors:** Navigates unfamiliar campus blocks, discovers administrative service desks, locates examination centers, and views landmark-guided walking directions.
2. **Persons with Disabilities (PWD) / Specially-Abled Users:** Receives strictly barrier-free, wheelchair-accessible routes utilizing ramps, designated elevators, and tactile paving while bypassing staircases and steep curbs.
3. **Faculty & Campus Staff:** Discovers optimized inter-departmental commute routes, finds scheduled meeting rooms, and broadcasts lecture hall relocations.
4. **Campus Estate & Event Administrator:** Manages digital spatial maps, defines temporary pathway closures (e.g., convocation road blockades), and pins live campus events or emergency help desks.

3. Key Functional Modules & Technical Deliverables

Module 1: Comprehensive Indoor-Outdoor Campus Mapping

- **Hybrid Indoor-Outdoor Graph Topology:** Vectorized multi-layered graph representation mapping outdoor walking paths, roadways, building entrance portals, corridors, stairwells, and elevators.
- **Multi-Floor Building Schematics:** Seamless layer switching between floors (Ground, Floor 1, Floor 2, Basement) with clear vertical transit nodes (elevators and stairs).
- **Point of Interest (POI) & Semantic Search:** Natural-language and category-based spatial search (e.g., "*Robotics Lab 304*", "*Central Library Reading Room*", "*Hostel Block B Mess*", "*Nearest ATM*").

Module 2: AI-Powered Context-Aware Pathfinding Engine

- **Multi-Criteria Route Optimization:** Computes tailored routing variations using graph algorithms (Dijkstra, A*, Hierarchical Pathfinding):
- **Shortest Path:** Minimal distance/time traversal.
- **Accessible / Barrier-Free Path:** Prioritizes elevators and ramps while strictly filtering out staircases and unpaved terrain.
- **Weather-Protected / Shaded Path:** Prefers covered corridors, canopy walkways, and indoor tunnels during extreme heat or heavy rainfall events.
- **Dynamic Obstacle & Blockade Rerouting:** Real-time graph weight recalculation to detour pedestrians around temporary campus construction, waterlogged alleys, or barricaded zones.

Module 3: Augmented Landmark Guidance & Visual Turn-by-Turn

- **Landmark-Anchored Guidance:** Supplements metric turn-by-turn prompts with recognizable campus cues (e.g., "*Turn left after the Central Fountain and take the glass elevator to the 3rd Floor*").
- **Indoor Positioning & Simulation:** Employs QR-code portal check-ins, simulated Wi-Fi/BLE beacon trilateration, or compass-guided step-counting (dead reckoning) for indoor tracking where GPS is unavailable.
- **Interactive First-Person / 2.5D Campus Perspective:** Clean visual map rendering with 3D building extrusions, directional route polylines, and step-by-step navigation cards.

Module 4: Campus Life Integration & Admin Geoportal

- **Live Event & Schedule Synchronization:** Connects with academic timetables and campus event calendars to auto-route students directly to their next scheduled lecture hall.
- **Administrative Map Studio:** Web-based editor allowing estate managers to draw, edit, or close paths, pin temporary kiosks, and update room allocations without redeploying code.

- **Emergency Safety & Security Layer:** Highlights one-tap navigation to the nearest campus security booths, emergency exits, fire extinguishers, first-aid infirmaries, and automated defibrillators (AEDs).

4. Recommended Technology Stack

Layer, Suggested Technologies

Frontend Web / Mobile, React.js (Vite) / Next.js / Tailwind CSS / Leaflet.js / Mapbox GL JS / OpenLayers / Three.js (2.5D/3D building rendering)

Mobile / PWA Client, "React Native, Flutter, or Progressive Web App (PWA) with HTML5 Geolocation & Device Orientation APIs"

Backend & Routing, Python (FastAPI / Flask) or Node.js (Express) with REST / WebSockets
Geospatial Database, "PostgreSQL with PostGIS / Supabase (Spatial indexing, GeoJSON geometries, ST_Distance, pgRouting)"

Graph & Path Planning, "NetworkX, custom Graph AI* engines, Turf.js for spatial vector calculations"

Mapping Standards, "GeoJSON, OpenStreetMap (OSM) tools (JOSM), indoor mapping schemas"

5. Evaluation & Judging Criteria (100 Points Total)

1. **Pathfinding Precision & Accessibility Logic (25 Points):** Demonstration of shortest, barrier-free (wheelchair/elevator), and context-aware routing with verified obstacle avoidance.
2. **Indoor-Outdoor Continuity & Multi-Floor UX (25 Points):** Smooth visual transitions between outdoor pathways and multi-level indoor building floorplans.
3. **Landmark-Driven Guidance & Usability (20 Points):** Quality of user-friendly landmark navigation cues, intuitive search, and mobile-first responsiveness.
4. **Admin Geo-Tools & Event Synchronization (15 Points):** Functionality of the administrative route-closure studio and dynamic campus life/event integration.
5. **Code Architecture, Sprint Execution & Standards (15 Points):** Clean GitHub repository, reproducible mock campus GeoJSON datasets, .env.example, and a working live interactive web demonstration.

6. Submission Guidelines

1. **Code Repository:** Public/Private GitHub repository containing clean branch commits, setup guide, mock campus GeoJSON datasets, and .env.example.
2. **Live Deployment:** Hosted interactive web application (Vercel / Render / Netlify / AWS).
3. **Demo Video / Deck:** A crisp 3-minute video showing:
4. Searching for an indoor campus destination across multiple floors.
5. Generating and comparing a "Shortest Path" vs. an "Accessible / Ramp-Only Route".
6. Dynamic path recalculation when an administrator sets an active pathway blockade.
7. Landmark-guided turn-by-turn navigation card walkthrough.